

Problem - 1,

Draw the graph of the function

$$F(x) = |x+1| + |x| + |x-1|$$

Given the function.

$$F(x) = |x+1| + |x| + |x-1|$$

$$= \begin{cases} -(x+1) - x - (x-1) ; & x < -1 \\ (x+1) - x - (x-1) ; & -1 \leq x < 0 \\ (x+1) + x - (x-1) ; & 0 \leq x < 1 \\ (x+1) + x + (x-1) ; & x \geq 1 \end{cases}$$

$$= \begin{cases} -x-1-x-x+1 & ; x < -1 \\ x+1-x-x+1 & ; -1 \leq x < 0 \\ 2x+1-x+1 & ; 0 \leq x < 1 \\ 3x & ; x \geq 1 \end{cases}$$

$$= \begin{cases} -3x & ; x < -1 \\ 2-x & ; -1 \leq x < 0 \\ x+2 & ; 0 \leq x < 1 \\ 3x & ; x \geq 1 \end{cases}$$

Now.

$$F(x) = -3x \quad ; \quad x < -1$$

x	-1	-2
$y = F(x)$	3	6

$$F(x) = 2 - x \quad ; \quad -1 \leq x < 0$$

x	-1	0
$y = F(x)$	3	2

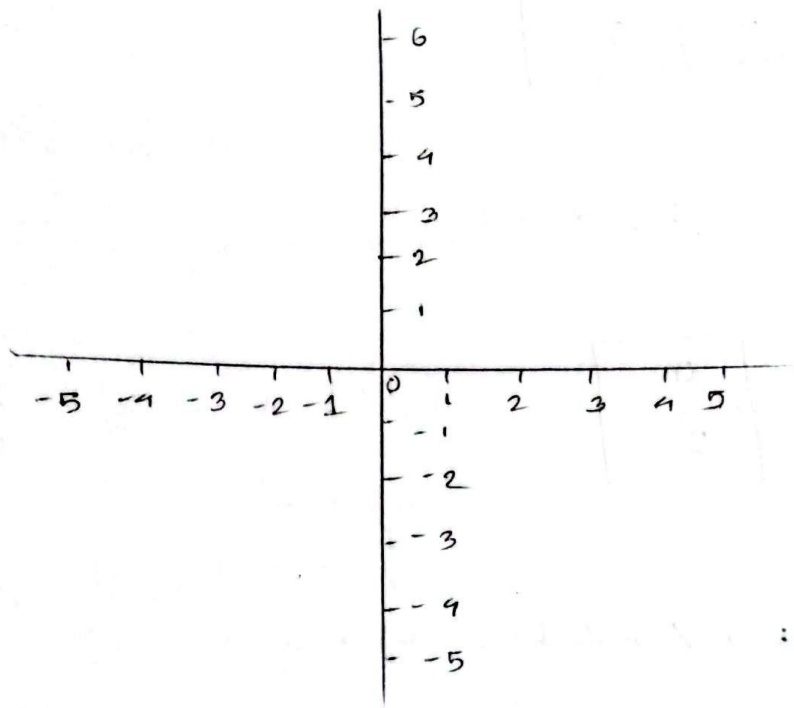
$$F(x) = x + 2 \quad ; \quad 0 \leq x < 1$$

x	0	1
$y = F(x)$	2	3

$$F(x) = 3x \quad ; \quad x \geq 1$$

x	1	2
$y = F(x)$	3	6

We sketch the following graph:



2	1	1
2	1	1

Problem :- Draw the graph of the function defined by:

$$F(x) = -|x-1| + |x+2| + |x-3|$$

Given the function,

$$F(x) = -|x-1| + |x+2| + |x-3|$$

$$= |x+2| - |x-1| + |x-3|$$

$$= \begin{cases} -(x+2) + (x-1) - (x-3) ; & x < -2 \\ (x+2) + (x-1) - (x-3) ; & -2 \leq x < 1 \\ (x+2) + (x-1) - (x-3) ; & 1 \leq x < 3 \\ (x+2) + (x-1) + (x-3) ; & x \geq 3 \end{cases}$$

$$= \begin{cases} -x-2 + x-1 - x+3 ; & x < -2 \\ x+2 + x-1 - x+3 ; & -2 \leq x < 1 \\ x+2 + x-1 - x+3 ; & 1 \leq x < 3 \\ x+2 - x+1 + x-3 ; & x \geq 3 \end{cases}$$

$$= \begin{cases} -x & ; x < -2 \\ x+4 & ; -2 \leq x < 1 \\ -x+6 & ; 1 \leq x < 3 \\ x & ; x \geq 3 \end{cases}$$

Now,

$$F(x) = -x \quad ; \quad x < -2$$

x	-2	-3
$y=F(x)$	2	3

$$F(x) = x+4 \quad ; \quad -2 \leq x < 1$$

x	-2	1
$y=F(x)$	2	5

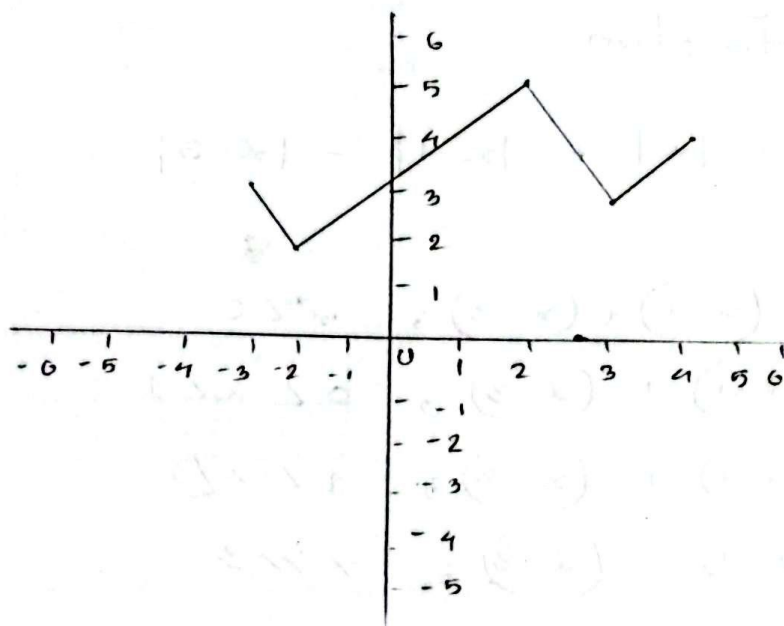
$$F(x) = -x+6 \quad ; \quad 1 \leq x < 3$$

x	1	3
$y=F(x)$	5	3

$$F(x) = x \quad ; \quad x \geq 3$$

x	3	4
$y=F(x)$	3	4

We sketch the following graph:-



Problem:- Draw the graph of the function

$$f(x) = |x| + |x-1| - |x-3|$$

Given the function.

$$f(x) = |x| + |x-1| - |x-3|$$

$$= \begin{cases} -x - (x-1) + (x-3); & x < 0 \\ x - (x-1) + (x-3); & 0 \leq x < 1 \\ x + (x-1) + (x-3); & 1 \leq x < 3 \\ x + (x-1) - (x-3); & x \geq 3 \end{cases}$$

$$= \begin{cases} -x - x + 1 + x - 3; & x < 0 \\ x - x + 1 + x - 3; & 0 \leq x < 1 \\ x + x - 1 + x - 3; & 1 \leq x < 3 \\ x + x - 1 - x + 3; & x \geq 3 \end{cases}$$

$$= \begin{cases} -x - 2; & x < 0 \\ x - 2; & 0 \leq x < 1 \\ 3x - 4; & 1 \leq x < 3 \\ x + 2; & x \geq 3 \end{cases}$$

Now.

$$f(x) = -x - 2 \quad ; \quad x < 0$$

x	0	-1
$y = f(x)$	-2	-1

$$f(x) = x - 2 \quad ; \quad 0 \leq x < 1$$

x	0	1
$y = f(x)$	-2	-1

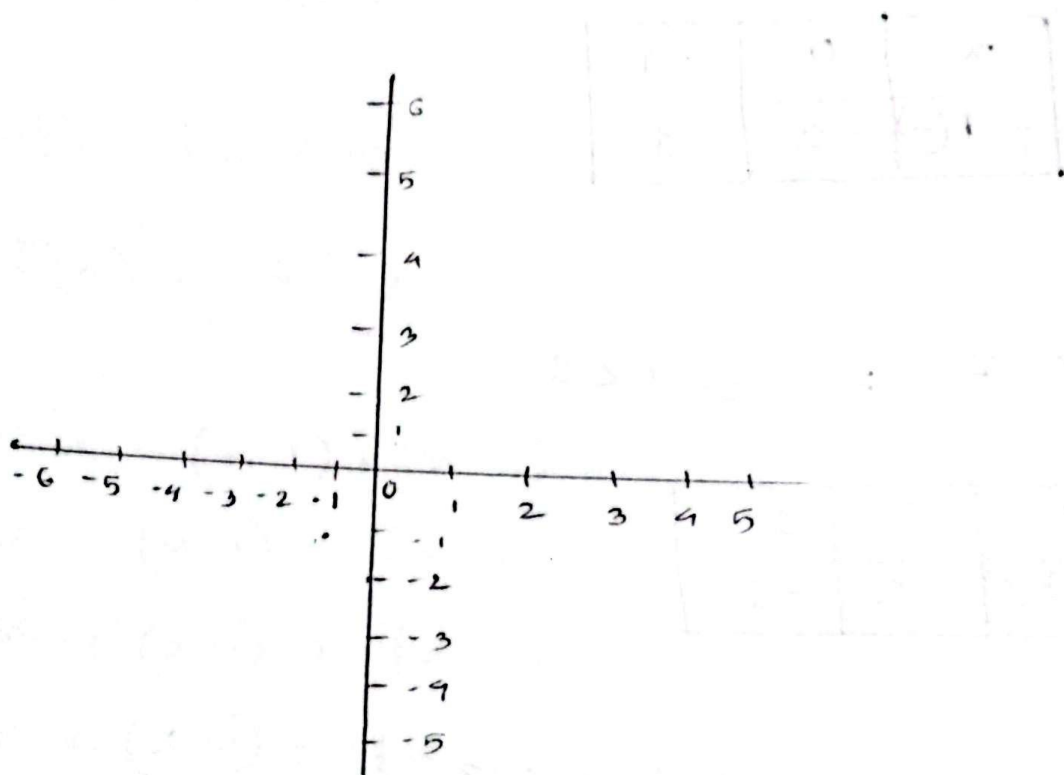
$$f(x) = 3x - 4 \quad ; \quad 1 \leq x < 3$$

x	1	3
$y = f(x)$	-1	5

$$f(x) = x + 2 \quad ; \quad x \geq 3$$

x	3	4
$y = f(x)$	5	6

we sketch the following graph:



Domain:-

Note:-

$$(i) 0 \leq x \leq 1 \rightarrow [0, 1]$$

$$(ii) 0 \leq x < 1 \rightarrow [0, 1)$$

$$(iii) 0 < x < 1 \rightarrow (0, 1)$$

$$(iv) 0 \leq x \leq 1 \rightarrow [0, 1]$$

$$* [0, 1) \cup [1, 3] \cup (3, 10]$$

$$= [0, 3] \cup (3, 10]$$

$$= [0, 10]$$

$$* [0, 1) \cup (1, 3]$$

$$= [0, 3] - \{1\}$$

* 'o' always 'open'

* $() \rightarrow$ open

* $[] \rightarrow$ close/include

Problem:- Find the domain and range for the following :-

$$f(x) = \begin{cases} x^2 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{when } x > 1 \end{cases}$$

$$\begin{aligned} \text{Domain} &= (-\infty, 0) \cup [0, 1] \cup (1, \infty) \\ &= (-\infty, 1) \cup (1, \infty) \\ &= (-\infty, \infty) \end{aligned}$$

(Ans)

For Range :- $f(x) = x^2$; when $x < 0$

x	$-\infty$	0
$f(x)$	∞	0

$f(x) = x$; when $0 \leq x \leq 1$

x	0	1
$f(x)$	0	1

$$F(x) = \frac{1}{x} \text{ , when } x > 1$$

x	1	∞
$F(x)$	1	0

$$\begin{aligned}
 \text{Range: } & (0, \infty) \cup [0, 1] \cup (0, 1) \\
 &= (0, \infty) \cup [0, 1] \\
 &= [0, \infty) \\
 &\quad (\text{Ans})
 \end{aligned}$$

Problem:- Find the domain and range for the following

$$F(x) = \begin{cases} 1+x & \text{when } -1 < x < 0 \\ 1-x & \text{when } 0 \leq x \leq 1 \\ x & \text{when } x > 1 \end{cases}$$